

Probing the Mechanisms of Chronotype Using Quantitative Modeling

A. J. K. Phillips,^{*,†,‡,1} P. Y. Chen,^{*,†} and P. A. Robinson^{*,‡}

^{*}*School of Physics, University of Sydney, New South Wales, Australia,*

[†]*Division of Sleep Medicine, Brigham & Women's Hospital, Harvard Medical School, Boston, Massachusetts,*

[‡]*Brain Dynamics Center, Sydney Medical School–Western, University of Sydney, Westmead, New South Wales, Australia*

Abstract The physiological mechanisms underlying interindividual differences in chronotype have yet to be established, although evidence suggests both circadian and homeostatic processes are involved. A physiologically based model is developed by combining models of the sleep-wake switch and circadian pacemaker, providing a means of examining how interactions between these systems affect chronotype. Specifically, chronotype is shown to depend on the relative influences of homeostatic and circadian drives, with a stronger homeostatic drive causing morningness. Changes to intrinsic circadian and homeostatic properties, including homeostatic clearance and production rates, and circadian period and amplitude, are also shown to affect chronotype. These results provide a framework for explaining several experimentally observed phenomena, including age-related morningness, adolescent eveningness, and familial advanced and delayed sleep-phase disorders. Additionally, experimental studies have shown that healthy adults on the extremes of the morningness-eveningness spectrum fall into two subtypes: those whose circadian phase markers are unaffected by chronotype, and those whose circadian phase markers track their chronotype. The model demonstrates that this spectrum likely results from interindividual differences in homeostatic kinetics in the first group, and differences in circadian period in the second group. Physiologically based modeling can thus guide diagnosis of sleep pathologies.

Key words sleep, mathematical model, chronotype, morningness, eveningness, physiology

1. INTRODUCTION

Chronotype is defined by preferred sleep-wake schedule, with evening types sleeping later than morning types (Kerkhof and Van Dongen, 1996). Wide interindividual variability is observed (Kerkhof, 1985), and in extreme cases can disrupt

social schedules and employment (Wittman et al., 2006). Age-related changes have been consistently reported, including adolescent eveningness (Carskadon et al., 1993, 1998), and morningness in old age (Monk et al., 1991). The physiological mechanisms underlying chronotype have yet to be fully elucidated. Some pathological chronotypes, including

1. To whom all correspondence should be addressed: Andrew J.K. Phillips, Division of Sleep Medicine, Brigham & Women's Hospital, 221 Longwood Ave., Suite 438, Boston, MA 02115; e-mail: ajphillips@partners.org.

familial advanced sleep-phase disorder (ASPD; Reid et al., 2001) and delayed sleep-phase disorder (DSPD; Weitzman et al., 1981) result from genetic mutations affecting circadian period, with longer periods causing eveningness (Toh et al., 2001, Archer et al., 2003; Brown et al., 2008). Yet in other cases, chronotype differences cannot be attributed to the circadian system, implicating sleep homeostasis (Mongrain and Dumont, 2007).

Human sleep is regulated by the ascending arousal system (AAS), a collection of brainstem and hypothalamic nuclei that diffusely innervate the cerebrum (Saper et al., 2005). Mutual inhibition between wake-active AAS monoaminergic nuclei (MA) and the sleep-active ventrolateral preoptic area (VLPO) forms the sleep-wake switch (Saper et al., 2001). Each group reinforces its own firing by inhibiting the other, yielding bistability.

Sleep timing is determined by homeostatic and circadian drives to the VLPO (Saper et al., 2005), which in turn receives behavioral feedback (Pace-Schott and Hobson, 2002). The circadian rhythm is maintained in the suprachiasmatic nucleus (SCN) and projected via the dorsomedial hypothalamus (DMH) to the VLPO (Gooley et al., 2006). The SCN is entrainable by photic (Czeisler et al., 1999) and nonphotic stimuli (Klerman et al., 1998), and maintains an approximately 24-h oscillation (Czeisler et al., 1999). The homeostatic sleep drive increases during wake and decreases during sleep. Its physiological basis remains unclear, but growing evidence implicates somnogen accumulation. Here, we model accumulation and clearance of basal forebrain adenosine (Porkka-Heiskanen et al., 2000), which inhibits basal forebrain neurons that inhibit the VLPO (Pace-Schott and Hobson, 2002). Cytokines also contribute, and the full pathway is unknown (Imeri and Opp, 2009).

To understand how chronotype differences arise, it is necessary to develop models that include circadian, homeostatic, and sleep-wake switch dynamics. It is insufficient to study these systems in isolation; as we will show, interactions have critical effects on the dynamics. Previously, human sleep has been modeled biomathematically (Achermann and Borbély, 2003; Van Dongen, 2004) and physiologically (Tamakawa et al., 2006; Phillips and Robinson, 2007; Rempé et al., 2009), while circadian modeling has developed largely separately (Jewett et al., 1999; Kronauer et al., 1999). Here we present a combined model of sleep physiology with a dynamic circadian oscillator.

Circadian models exist at various levels (Beersma, 2005), ranging from molecular (Leloup

and Goldbeter, 2003), to single (Kronauer et al., 1999) or coupled oscillators (Kronauer et al., 1982; Leise and Siegelmann, 2006). However, interactions between the circadian system and sleep-wake state have not been extensively explored. The most obvious interaction is modulation of light by sleeping times, which has been considered previously (Beersma et al., 1987), but not widely adopted. Recently, imposed sleep times were used to gate incoming light signals (St. Hilaire et al., 2007), but the effect of the circadian drive on sleep times was not modeled, which is necessary to predict chronotype. We implement this here.

In Section 2, we construct a quantitative model by coupling a model of the sleep-wake switch, with a circadian oscillator model. This combined model is then used to explore the effects of variations of physiological parameters on chronotype in Section 3, yielding new insights into the underlying mechanisms. We focus on parameters associated with the circadian and homeostatic drives, since these are most likely to affect chronotype. These results are shown to help in diagnosis of chronotype disorders by inferring the pathology from observed behavior. Broader implications and future directions are discussed in Section 4.

2. MATERIALS AND METHODS

We introduce the model of Phillips and Robinson (2007) in Section 2.1, before introducing the circadian oscillator model of Forger et al. (1999) in Section 2.2. The models are combined in Section 2.3.

2.1 Sleep-Wake Switch Model

To model the AAS and homeostatic drive, the Phillips-Robinson (2007) model uses average properties ascribed to neuronal populations (Nunez, 1995). The model includes the mutually inhibitory MA and VLPO populations. Although subsidiary pathways may exist, circadian (C) and homeostatic (H) inputs drive only the VLPO for simplicity, while MA receives an averaged input A from cholinergic and orexinergic sources. For each population, $j = v, m$, where v is VLPO, and m is MA, we define mean cell body potential $V_j(t)$ relative to resting, and mean firing rate $Q_j(t)$. Because of their small sizes, we assume spatial homogeneity of each population, and neglect propagation delays. Firing rate is approximated by the sigmoid,

Table 1. Nominal parameter values for the Phillips-Robinson (2008) model (left side) and the Forger et al. (1999) model (right side).

Parameter	Value	Unit	Parameter	Value	Unit
v_{vc}	-5.8	mV	λ	60	h^{-1}
v_{vh}	1.0	mV nM^{-1}	α_0	0.16	1
χ	45	h	I_0	9500	lux
μ	4.4	nM s	p	0.6	1
D_0	-10.2	mV	β	0.013	1
Q_{max}	100	S^{-1}	b	0.4	1
θ	10	mV	G	19.9	1
σ'	3	mV	k	0.55	1
A	1.3	mV	τ_c	24.2	h
v_{vm}	-2.1	mV s	f	0.99669	1
v_{mv}	-1.8	mV s	γ	0.23	1
τ_m, τ_v	10	s			

$$Q_j(t) = S[V_j(t)] = \frac{Q_{max}}{1 + \exp[(\theta - V_j)/\sigma']} \quad (1)$$

where Q_{max} is the maximum possible rate, θ is mean firing threshold, and $\sigma'\pi/\sqrt{3}$ is the standard deviation of θ (Robinson et al., 1997).

The model differential equations are:

$$\tau_v \frac{dV_v}{dt} + V_v = v_{vm}Q_m + D \quad (2)$$

$$\tau_m \frac{dV_m}{dt} + V_m = v_{mv}Q_v + A \quad (3)$$

where v_{jk} terms weight input to population j from k , τ_j is the characteristic decay time for the neuromodulator expressed by group j , and total sleep drive is:

$$D = v_{vc}C + v_{vh}H + D_0 \quad (4)$$

where D_0 is a constant offset. The circadian drive C has previously been approximated by a sinusoid, but is modeled using an oscillator here, as described below. The homeostatic drive H represents extracellular basal forebrain adenosine concentration, with its form motivated by the two-process model (Borbély, 1982). Other somnogens may play a role, and the same functional form is equally applicable to other factors that accumulate during wake and clear during sleep. Clearance rate is assumed to be proportional to concentration, and production is approximated as

a linear function of Q_m , since MA activity correlates with arousal (Aston-Jones et al., 1991), yielding:

$$\chi \frac{dH}{dt} + H = \mu Q_m \quad (5)$$

where χ is the homeostatic time constant for clearance and accumulation, and μ is constant. Parameter values have previously been constrained under the assumption of entrainment (Phillips and Robinson, 2007, 2008), meaning their values should need little refinement when combined with the circadian model.

We thus use the parameter values of Table 1 except where otherwise stated.

2.2 Forger et al. Model

To model the circadian rhythm we use a simplified version of the Jewett-Kronauer oscillator model (Forger et al., 1999). As seen in Figure 1A the model includes two components: 1) retinal processing of photic stimuli (Process L), and 2) the pacemaker (Process P).

Photoreceptors are converted from ready to activated by photons, at a rate α , dependent on the light intensity that reaches the retina, $\tilde{I}$. Activated photoreceptors are converted back to ready at a constant rate β . The fraction n of photoreceptors that are activated thus obeys:

$$\frac{dn}{dt} = \lambda[\alpha(1 - n) - \beta n] \quad (6)$$

where $1/\lambda$ is the receptor time constant, and $\alpha = \alpha_0(\tilde{I}/I_0)^p$ (Kronauer et al., 1999).

The resultant photic drive B to the pacemaker is then modeled as proportional to the rate $\alpha(1 - n)$ at which photoreceptors are activated, giving:

$$B = (1 - bx)(1 - by)\hat{B} \quad (7)$$

$$\hat{B} = G\alpha(1 - n) \quad (8)$$

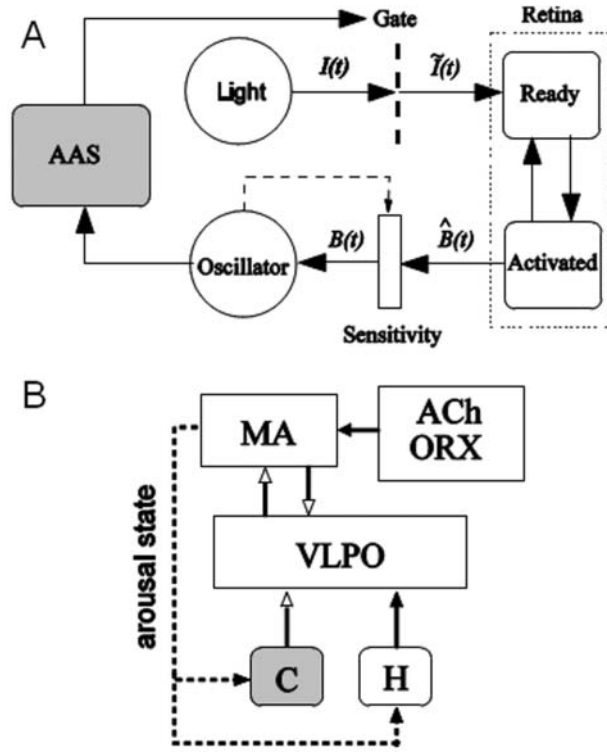


Figure 1. (A) The circadian model and its connection to the ascending arousal system (AAS). Environmental light $I(t)$ is gated by arousal state (thick dashed line) to yield light incident on the retina, $\tilde{I}(t)$, activating receptors (Process L). Retinal output $\hat{B}(t)$ is modulated by circadian phase to yield the drive $B(t)$, to the pacemaker (Process P). (B) Mutual inhibition between MA and VLPO forms the sleep-wake switch, with circadian (C) and homeostatic (H) drives to VLPO. Cholinergic and orexinergic drives (ACh/ORX) excite MA, and arousal state interacts with C and H.

where b and G are constants. The term $(1 - bx)(1 - by)$ is included to account for the experimentally observed circadian phase dependence of light sensitivity (Kronauer et al., 1999).

The pacemaker is then modeled by 2 first order differential equations, based on a van der Pol oscillator:

$$\kappa \frac{dx}{dt} = \gamma \left(x - \frac{4x^3}{3} \right) - y \left[\left(\frac{24}{f\tau_c} \right)^2 + kB \right] \quad (9)$$

$$\kappa \frac{dy}{dt} = x + B \quad (10)$$

where y represents pacemaker activity, and x is an auxiliary variable. The parameter τ_c is intrinsic period, k is constant, f is included to fix the period at τ_c , γ is the stiffness of the oscillator, and $\kappa = (12/\pi)$ h (Forger et al., 1999). This model has been previously calibrated to reproduce human entrainment dynamics, including the phase-response curve (Kronauer et al., 1999).

2.3 Combined Model

In combining the 2 models, we assume the circadian signal reaching the VLPO is a linear function of y , as a first approximation, since this variable is correctly phased. Because y is approximately sinusoidal and SCN firing rate is close to zero during biological night, we use $C = (1 + y)/2$, rescaling v_{vc} (shown rescaled in Table 1) for consistency with previous work. One can interpret C as the drive relayed by the DMH. Since this connection is GABAergic, we expect $v_{vc} < 0$ mV. A schematic of the combined model is shown in Figure 1.

An additional link between the models is gating of environmental light by arousal state. This feedback has previously appeared in phenomenological composite fatigue models (Mallis et al., 2004). To simulate eye closure during sleep (and subjects seeking dimmer conditions), throughout Section 3 we use:

$$\tilde{I}(t) = \Theta(Q_m - Q_m^{th}) I(t) \quad (11)$$

where $I(t)$ is ungated intensity, Θ is a step function, and $Q_m^{th} = 1 \text{ s}^{-1}$ is a threshold firing rate above which the state is defined as wake, based on experiment (Aston-Jones and Bloom, 1981). For Sections 3.1 to 3.3, we use:

$$I(t) = I_A (1 + \cos \omega t) / 2 \quad (12)$$

to simulate a 24-h environmental light schedule, where $I_A = 10^4$ lux, and $t = 0$ corresponds to 1200 h. While this is a strong light intensity, even room light is a comparably strong zeitgeber because of saturating retinal response at high intensities. In Section 3.4, we use light data reported in Goulet et al. (2007). During sleep, retinal light intensity is reduced to $I = 0$ by the gate, and this, in turn, results in a 3-h phase delay of the circadian oscillator relative to $I(t)$, causing it to peak in activity around 1500 h. Without this gating, the circadian oscillator entrains to the same phase as $I(t)$, peaking at 1200 h, which is inconsistent with experiment. It is shown in Section 3 that inclusion of gating has significant influence on phase-related dynamics, and thus chronotype.

To analyze model dynamics, we precisely define phase using the Hilbert transform (Brychkov and Prudnikov, 1989), which extracts time-varying phase and amplitude of an approximately sinusoidal signal. The instantaneous phase $\phi_X(t)$ of any variable $X(t)$ is given by:

$$\phi_X(t) = \tan^{-1} [H(X - \bar{X}) / (X - \bar{X})] \quad (13)$$

where $\bar{X}$ is the mean of X and H is the Hilbert Transform, defined by:

$$H[X(t)] = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{X(\tau)d\tau}{t - \tau}. \quad (14)$$

For $X(t) = \cos t$, $H[X(t)] = \sin t$, so the phase is $\tan^{-1}(\sin t / \cos t) = t$. The same method can be applied to approximately sinusoidal functions, such as C and I . Under entrained conditions, phase increases at a mean rate of 2π radians per day.

The phase difference between variables X and Y is denoted by $\Delta\phi_{XY} = \phi_X - \phi_Y$. Throughout Section 3, we use $\Delta\phi_{CI}$ as a measure of circadian phase, and $\Delta\phi_{DI}$ as a measure of sleep phase. When comparing with experiment in Sections 3.1.2 and 3.4, we use dim light melatonin onset (DLMO) as a measure of circadian phase, and the midpoint of habitual sleep periods as a measure of sleep phase. To estimate circadian amplitude, we use core body temperature (CBT) measures.

During entrainment, $\Delta\phi_{CI}$ is approximately -0.8 radians, corresponding to C lagging approximately 3 h behind I . In simulations, slight daily variations (< 0.1 radians) in $\Delta\phi_{CI}$ occur as the circadian phase advances and delays relative to I due to light exposure being distributed across different portions of the model's phase-response curve. Average values of phase differences are thus used in Section 3, and to generate the results, the model is run long enough to ensure entrainment (requiring that $\Delta\phi_{DI}$ changes < 5 sec between days). If entrainment was not achieved within 500 simulated days, then it was considered to have failed, with this long time limit being used to exclude cases where the model appeared to entrain at first but gradually slipped out of phase. Matlab was used for all simulations, and since we are interested in average differences in phase, no noise was added.

3. RESULTS

We now use the combined model to investigate the physiological causes of differences in chronotype. We begin by examining how chronotype depends on relative strengths of circadian and homeostatic drives in Section 3.1, including comparison with experiment for age-related morningness. We then examine how chronotype is affected by interindividual differences in circadian period in Section 3.2, and homeostatic kinetics in Section 3.3. In Section 3.4, we apply the model to the experiments of Goulet et al. (2007),

showing that it enables determination of the physiological causes of chronotype differences from data.

3.1 Relative Drive Strengths

The timing of the sleep-wake cycle is determined by the combined influences of homeostatic and circadian processes, represented by total sleep drive D in our model. The phase $\Delta\phi_{DI}$ of D , relative to the environmental light I , is therefore dependent on the relative strengths of circadian and homeostatic signals, that is, on v_{vc} and v_{vh} . With nominal parameters, the circadian drive to sleep is maximum around 0400 h, whereas H peaks at the end of waking. The phase of D is therefore in between, with a relative increase in homeostatic drive strength (e.g., by increasing v_{vh} or decreasing v_{vc}) generally resulting in phase advance, that is, morningness. However, as shown below, the situation is more complex than this, because such changes also alter light exposure. In Section 3.1.1, we examine the effects of changing v_{vh} . Then in Section 3.1.2, we examine the effects of changing circadian amplitude, and relate this to age-related morningness.

3.1.1 Homeostatic Drive Strength

In Figure 2, sleep phase is shown as a function of v_{vh} . Increasing v_{vh} causes morningness, but also increases sleep duration. Once sleep duration becomes too long, there is insufficient light exposure to entrain the oscillator, possibly because of the circadian model not being calibrated to accurately model short photoperiods. Decreasing v_{vh} initially causes a shift toward eveningness, and decreases sleep duration. As $v_{vh} \rightarrow 0$ reduced sleep duration becomes the dominant effect and for $v_{vh} < 0.6$ mV nM $^{-1}$ the model predicts a pathological state of permanent wakefulness, implying that normally $v_{vh} > 0.6$ mV nM $^{-1}$. With no sleep to gate incoming light, the circadian oscillator entrains to the same phase as I , causing morningness. We note that since D oscillates, we can define phase even without sleep.

3.1.2 Circadian Amplitude and Age-Related Morningness

Experiments indicate that the amplitude of circadian CBT (Czeisler et al., 1992; Monk et al., 1995; Dijk et al., 2000), melatonin (Duffy et al., 1998; Pandi-Perumal et al., 2005), and sleep-wake rhythms

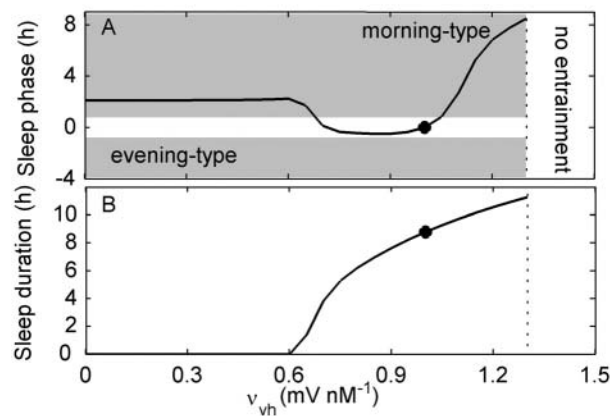


Figure 2. (A) Sleep phase relative to nominal (using Table 1 values), with positive values indicating advance, and (B) sleep duration, as functions of homeostatic drive strength v_{vh} . Nominal values are indicated by heavy dots, and phases differing by more than an hour from nominal are arbitrarily labeled by shaded regions, corresponding to severe morning and evening chronotypes. Entrainment fails beyond the dotted line.

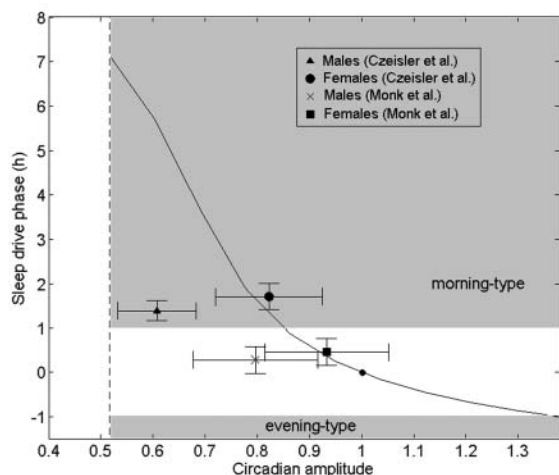


Figure 3. Sleep phase relative to nominal as a function of circadian amplitude (normalized value: $v_{vc}/-5.9$ mV). The nominal value is plotted as a heavy dot, with data (Czeisler et al., 1992; Monk et al., 1995) for the elderly, as labeled. Error bars indicate SEM.

(Ceolim et al., 1996; Münch et al., 2005) decline with age, whereas period seems unaffected (Czeisler et al., 1999). These changes may be due to SCN decay (Bliwise, 2000)—one study showed a 50% decrease in volume with age (Swaab et al., 1985)—and are coincident with morningness (Monk et al., 1991; Dijk et al., 2000). We model declining circadian amplitude by

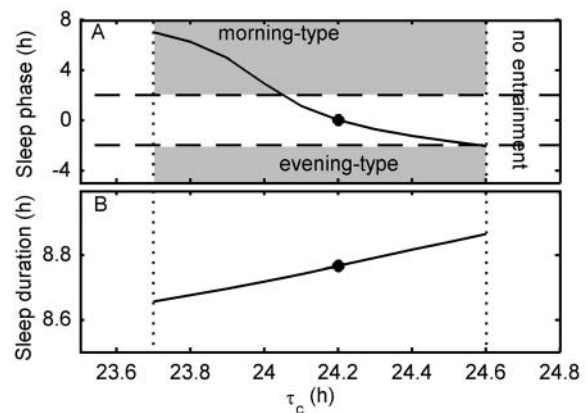


Figure 4. (A) Sleep phase relative to nominal, and (B) sleep duration, as functions of free-running period, τ_c . The nominal value is plotted as a heavy dot. Entrainment fails outside the dotted lines.

varying v_{vc} . As shown in Figure 3, decreasing v_{vc} results in morningness. Sleep duration is barely affected by v_{vc} , indicating that reduced sleep in the elderly may be due to inability to maintain consolidated sleep, rather than a lessened intrinsic drive (Klerman and Dijk, 2008). Entrainment fails altogether for $v_{vc} > -1.3$ mV, indicating a physiological lower bound on amplitude. We compare the model with experimental data for males and females from 2 studies (Czeisler et al., 1992; Monk et al., 1995), using midpoint of the habitual sleep period to estimate sleep phase, and CBT to infer circadian amplitude. Normalized amplitudes are calculated by dividing elderly values by young values. Intriguingly, the model quantitatively reproduces the data for females but not males. This suggests aging differentially affects the SCN between sexes, implications of which are pursued in Section 4.

3.2 Circadian Period

Intrinsic circadian period averages 24.2 h across the human population (Czeisler et al., 1999), but mutations in clock genes can significantly alter this, resulting in pathological chronotypes (Brown et al., 2008). Modeling this by varying τ_c reproduces these phenomena, with increased τ_c resulting in eveningness, and decreased τ_c resulting in morningness, shown in Figure 4A. Furthermore, the oscillator is only entrainable for $23.7 \text{ h} < \tau_c < 24.6 \text{ h}$ using this light curve. As shown in Figure 4B, changes to τ_c have little effect on sleep duration, despite the strong effect on chronotype.

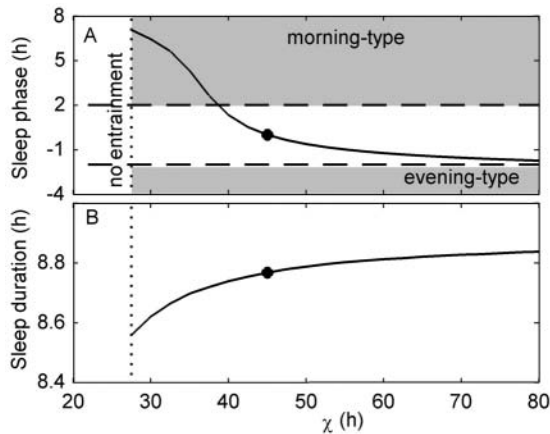


Figure 5. (A) Sleep phase relative to nominal, and (B) sleep duration, as functions of homeostatic time constant, χ . The nominal value is plotted as a heavy dot. Entrainment fails outside the dotted line.

3.3 Homeostatic Drive Parameters

One might expect chronotype to depend on only circadian variables, and until relatively recently this was the prevailing view. However, because sleep times are dictated by both homeostatic and circadian drives, homeostatic parameters must also contribute. The advantage of modeling is quantitative predictions of the effects of interindividual differences in homeostatic kinetics.

Homeostatic production is controlled by μ in equation (5). Changing this parameter has similar effects to changing v_{vh} in Section 3.1.1: increasing μ results in morningness and increased sleep duration, whereas decreasing μ results in eveningness and eventually a pathological sleepless state.

The homeostatic time constant has been estimated to satisfy $34 \text{ h} < \chi < 72 \text{ h}$ for normal adults, based on recovery from total sleep deprivation (Phillips and Robinson, 2008). Interindividual differences in homeostatic rise and decay rates have been identified as a possible source of chronotype differences (Taillard et al., 2003; Mongrain and Dumont, 2007), and adolescent eveningness (Carskadon et al., 1998; Jenni et al., 2005). Figure 5 shows that increasing χ results in a shift toward eveningness and slightly increased sleep duration, while decreasing it results in morningness.

3.4 Diagnosis of Chronotype Disorders

We now use the findings of the previous sections—summarized in Table 2—to motivate use of

the model as a diagnostic tool for chronotype disorders. In 2004, it was reported (Mongrain et al., 2004) that a selection of morning types and evening types could be separated into those whose circadian rhythms did not track their chronotype (“Intermediates”) and those whose did (“Extremes”), which suggested 2 independent mechanisms.

In a follow-up study Goulet et al. (2007) recorded light exposures for Intermediates and Extremes, providing sufficient information for us to accurately model the experimental scenario. In both groups, light exposure was low during sleep periods. In Intermediates, light exposure differed little between morning types and evening types, with both groups receiving most exposure from 1000 h to 2000 h. Timing of DLMO also did not differ, suggesting that circadian pacemaker properties are unchanged between the groups, for example, τ_c . Additionally, morning types slept about 45 min less than evening types. Referring to Table 2, χ is the only parameter whose variation increases morningness and decreases sleep duration. These results support the hypothesis that those in the Intermediate group differ in their position on the morningness-eveningness spectrum due to differences in χ , with Intermediate morning types having smaller χ . Simulating the model protocol (including light exposures), we plot model predictions of circadian and sleep phase alongside experimental data in Figure 6. As predicted, Intermediates lie along the axis in this 2-dimensional space predicted by variations in χ .

For Extremes, evening types receiving most light input approximately 2 h later than morning types. Furthermore, evening types expressed DLMO 5 h later than morning types. This, coupled with the fact that sleep duration did not differ significantly between groups, suggested that circadian oscillator properties differ (Mongrain et al., 2004), with morning types having a shorter period. Modeling this, we find that Extremes lie along the axis in the 2-dimensional space predicted by variations in τ_c shown in Figure 6.

4. DISCUSSION

We have developed a combined model of sleep physiology and circadian rhythms and used it to investigate physiological mechanisms responsible for interindividual differences in chronotype. Previously, identifying the cause of chronotype differences was a complicated exercise, with experiments variously implicating circadian period (Duffy et al., 1998) and

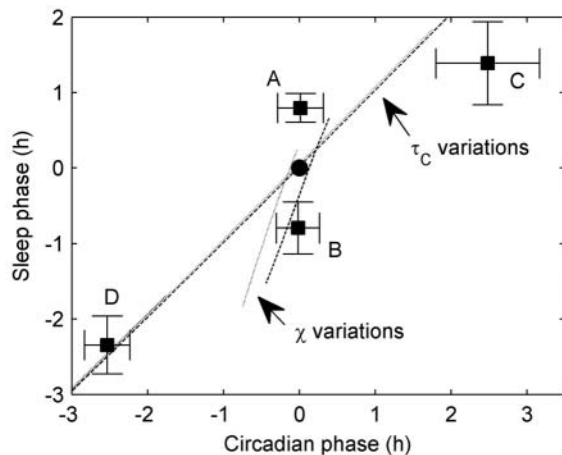


Figure 6. Sleep phase relative to normal, versus circadian phase relative to normal, showing data (Goulet et al., 2007) for: A) Intermediate morning types, B) Intermediate evening types, C) Extreme morning types, D) Extreme evening types. Error bars indicate SEMs, and the normal value (heavy dot) is the average of A and B. Short lines show model predictions for $30 \text{ h} < \chi < 72 \text{ h}$ (spanning the healthy range determined in Phillips and Robinson, 2008), using the light exposures reported for A (dashed) and B (dotted). Long lines show model predictions for $23.8 \text{ h} < \tau_c < 24.6 \text{ h}$ (spanning the range of entrainment), using the light exposures reported for C (dashed) and D (dotted).

homeostatic kinetics (Taillard et al., 2003; Mongrain and Dumont, 2007). More recently, subtypes were identified within the 1-dimensional morningness-eveningness spectrum (Goulet et al., 2007). Our model provides a means to quantitatively explore how homeostatic and circadian processes interact to yield chronotype.

Sleep timing results from competition between circadian and homeostatic drives, as shown in Section 3.1, because these drives peak at different phases. Increasing homeostatic drive strength causes morningness, as does decreasing circadian amplitude. Interactions between these processes were shown to be critical, underlining the importance of modeling both components in understanding chronotype. For example, changes in v_{vh} resulted in changes in sleep duration, which in turn modified light exposure and circadian phase.

In Section 3.1.2, we modeled aging by decreasing circadian amplitude, which resulted in morningness, consistent with experimental findings (Monk et al., 1991; Dijk et al., 2000). Interestingly, the model accurately reproduced chronotype for old females (Czeisler et al., 1992; Monk et al., 1995), whereas old males exhibited less change in chronotype than predicted, given their much larger amplitude reduction,

Table 2. Effects of increasing each parameter on morningness and sleep duration.

Parameter	Morningness	Sleep Duration
$-v_{vc}$	↑	=
$-v_{vh}$	↑	↑
$-\tau_c$	↑	=
μ	↑	↑
$-\chi$	↑	↓

↑ = increase; ↓ = decrease; "=" = no significant change.

suggesting a secondary mechanism may be involved. This is particularly noteworthy because male subjects experience more rapid decline of circadian amplitude with age (Monk, 2005), and the male SCN is structurally different from the female SCN (Swaab et al., 1985; Swaab et al., 2003). It may be that as the SCN loses volume, the individual cells desynchronize, and that this process is different in males. We also found that sleep duration is not significantly affected by amplitude. Experiments are inconsistent on this point, with some showing sleep duration decreases with age (Bixler et al., 1984), and others finding no changes (Tune, 1968). The view that sleep need does not change but sleep structure and ability to maintain sleep does (Klerman and Dijk, 2008) may be closer to the truth, with VLPO deterioration likely playing an important role, and care must be taken in separating healthy from pathological aging (Bliwise, 2000). Future physiological studies may help to direct development of pharmaceuticals to combat age-related decline in sleep quality, for example, targeting the dorsomedial hypothalamus to amplify the circadian signal.

In Section 3.2 increasing τ_c was shown to cause eveningness, as the oscillator lagged behind the day, while decreasing τ_c caused morningness. Experimentally, genetic mutations have been shown to result in abnormal circadian periods (Toh et al., 2001; Brown et al., 2008), manifesting as ASPD (Reid et al., 2001) or DSPD (Weitzman et al., 1981). Independent of genetic studies, it has been shown that τ_c correlates with chronotype (Duffy et al., 1998; Duffy et al., 2001). Model predictions are consistent with these results, and additionally predict that if τ_c and day length differ by too much then entrainment fails, as also confirmed by forced desynchrony experiments (Czeisler et al., 1986). Interestingly, some experiments have shown plasticity in human circadian period in response to changes in imposed day length (Scheer

et al., 2007), which implies the existence of a behavioral component to chronotype. For example, repeated evening light exposure may increase τ_c , reinforcing eveningness. Photoperiod exposure during early development may also contribute to chronotype (Mongrain et al., 2006). We have not attempted to model these phenomena here, but they should be examined in the future.

In Section 3.3, we showed that interindividual differences in homeostatic kinetics also affect chronotype. A longer homeostatic time constant resulted in eveningness, consistent with suggestions this may be the mechanism behind adolescent eveningness (Carskadon et al., 1993, 1998; Roenneberg et al., 2004; Jenni et al., 2005). Experiments indicate τ_c is no longer in adolescent evening types than the general population (Carskadon et al., 1999), and confirm that homeostatic kinetics are slower in mature adolescents than prepubescent adolescents (Jenni et al., 2005). Changes in χ may also contribute to age-related morningness (Dijk et al., 2000), but additional data are needed to quantify this effect and any sex-related differences. At present, the physiological pathway and the relative importance of adenosine accumulation for the homeostatic drive is not known, but physiologically based modeling may help to elaborate it.

The results of Sections 3.1 to 3.3 illustrate that there are many possible mechanisms underlying pathological chronotypes. In Section 3.4, we demonstrated that given appropriate data—sleep times, DLMO, and light exposure—the model can be used as a diagnostic tool to draw inferences from data regarding underlying physiology. Within the broader classifications of “morning type” and “evening type” we confirmed subtypes associated with changes in either homeostatic time constant or circadian period. Different physiological causes of altered chronotype differentially affect circadian phase, providing a metric to classify individuals. It seems very likely that previous experiments have included a mix of these subpopulations, resulting in studies variously finding group averaged correlations of both τ_c (Duffy et al., 1998) and χ (Mongrain and Dumont, 2007) with chronotype. Furthermore, we have not ruled out the possibility of correlations between these mechanisms.

Attributing physiological causes for interindividual differences in chronotype not only provides a means of individualizing treatments, but also validates disorders that lack wide public acceptance and can be dismissed as character flaws. With most of society running on the same schedule, abnormal chronotypes face social, school, and employment

problems (Wittman et al., 2006). Experiments indicate that chronotype is an inherent trait that persists despite imposed schedules (Carskadon et al., 1998).

By integrating circadian and noncircadian models, we have shown that it is possible to delineate and quantify relationships between physiology and chronotype. In the future, more elaborate circadian models should be used, for example, that of St. Hilaire et al. (2007). The Forger et al. (1999) model has limitations in protocols involving sharp amplitude recovery, such as type 0 resetting (Indic et al., 2005), but should not significantly affect our results, since no such resetting occurs here. Direct effects of light on the VLPO (Lu et al., 1999) have been omitted, and some evidence suggests the circadian system directly interacts with the homeostatic process and arousal state (Deboer et al., 2003; Schmidt et al., 2009), which should be considered in future work. Other applications include forced and spontaneous desynchrony protocols, which will enable improved calibration of phase offsets between the sleep-wake cycle and the circadian rhythm.

ACKNOWLEDGMENTS

This work was supported by the Australian Research Council, the National Health and Medical Research Council, and the National Space Biomedical Research Institute through NASA NCC 9-58.

REFERENCES

- Achermann P and Borbély AA (2003) Mathematical models of sleep regulation. *Front Biosci* 8:683-693.
- Archer SN, Robilliard DL, Skene DJ, Smits M, Williams A, Arendt J, and von Schantz M (2003) A length polymorphism in the circadian clock gene *Per3* is linked to delayed sleep phase syndrome and extreme diurnal preference. *Sleep* 26:413-415.
- Aston-Jones G and Bloom FE (1981) Activity of norepinephrine-containing locus coeruleus neurons in behaving rats anticipates fluctuations in the sleep-waking cycle. *J Neurosci* 1:876-886.
- Aston-Jones G, Chiang C, and Alexinsky T (1991) Discharge of noradrenergic locus coeruleus neurons in behaving rats and monkeys suggests a role in vigilance. *Prog Brain Res* 88:501-520.
- Beersma DGM, Daan S, and Dijk DJ (1987) Sleep intensity and timing: A model for their circadian control. In *Lectures on Mathematics in the Life Sciences Vol. 19: Some Mathematical Questions in Biology: Circadian Rhythms*, Carpenter GA, ed. Providence, RI: American Mathematical Society.
- Beersma DGM (2005) Why and how do we model circadian rhythms? *J Biol Rhythms* 20:304-313.

- Bixler EO, Kales A, Jacoby JA, Soldatos CR, and Vela-Bueno A (1984) Nocturnal sleep and wakefulness: Effects of age and sex in normal sleepers. *Int J Neurosci* 23:33-42.
- Bliwise DL (2000) Normal aging. In *Principles and Practice of Sleep Medicine*, Kyger MH, Roth T, and Dement WC, eds, 3rd ed, Ch. 3, pp 26-42. Philadelphia: Saunders.
- Borbély AA (1982) A two process model of sleep regulation. *Hum Neurobiol* 1:195-204.
- Brown SA, Kunz D, Dumas A, Westermarck PA, Vanselow K, Tilmann-Wahnschaffe A, Herzel H, and Kramer A (2008) Molecular insights into human daily behavior. *Proc Natl Acad Sci U S A* 105:1602-1607.
- Brychkov YA and Prudnikov AP (1989) *Integral Transforms of Generalized Functions*. New York: Gordon and Breach.
- Carskadon MA, Vieira C, and Acebo C (1993) Association between puberty and delayed phase preference. *Sleep* 16:258-262.
- Carskadon MA, Wolfson AR, Acebo C, Tzischinsky O, and Seifer R (1998) Adolescent sleep patterns, circadian timing, and sleepiness at a transition to early school days. *Sleep* 21:871-881.
- Carskadon MA, Labyak SE, Acebo C, and Seifer R (1999) Intrinsic circadian period of adolescent humans measured in conditions of forced desynchrony. *Neurosci Lett* 260:129-132.
- Ceolim MF, Campedelli MC, and Menna-Barreto L (1996) Circadian amplitude and quality of sleep in a group of active elderly subjects. *Biol Rhythm Res* 27:398-408.
- Czeisler CA, Allan JS, and Kronauer RE (1986) A method to assess the intrinsic period of the endogenous circadian oscillator. *Sleep Res* 15:266.
- Czeisler CA, Dumont M, Duffy JF, Steinberg JD, Richardson GS, Brown EN, Sanchez R, Rios CD, and Ronda JM (1992) Association of sleep-wake habits in older people with changes in output of circadian pacemaker. *Lancet* 340:933-936.
- Czeisler CA, Duffy JF, Shanahan TL, Brown EN, Mitchell JF, Rimmer DW, Ronda JM, Silva EJ, Allan JS, Emens JS, et al (1999) Stability, precision, and near-24-hour period of the human circadian pacemaker. *Science* 284:2177-2181.
- Deboer T, Vansteensel MJ, Détári L, and Meijer JH (2003) Sleep states alter activity of suprachiasmatic nucleus neurons. *Nat Neurosci* 6:1086-1090.
- Dijk DJ, Duffy JF, and Czeisler CA (2000) Contribution of circadian physiology and sleep homeostasis to age-related changes in human sleep. *Chronobiol Int* 17:285-311.
- Duffy JF, Dijk DJ, Klerman EB, and Czeisler CA (1998) Later endogenous circadian temperature nadir relative to an earlier wake time in older people. *Am J Physiol* 275:R1478-R1487.
- Duffy JF, Rimmer DW, Czeisler CA (2001) Association of intrinsic circadian period with morningness-eveningness, usual wake time, and circadian phase. *Behav Neurosci* 115:895-899.
- Forger DB, Jewett ME, and Kronauer RE (1999) A simpler model of the human circadian pacemaker. *J Biol Rhythms* 14:532-537.
- Gooley JJ, Schomer A, and Saper CB (2006) The dorsomedial hypothalamic nucleus is critical for the expression of food-entrainable circadian rhythms. *Nat Neurosci* 9:398-407.
- Goulet G, Mongrain V, Desrosiers C, Paquet J, and Dumont M (2007) Daily light exposure in morning-type and evening-type individuals. *J Biol Rhythms* 22:151-158.
- Imeri L and Opp MR (2009) How (and why) the immune system makes us sleep. *Nat Rev Neurosci* 10:199-210.
- Indic P, Forger DB, St. Hilaire MA, Dean DA, Brown EN, Kronauer RE, Klerman EB, and Jewett ME (2005) Comparison of amplitude recovery dynamics of two limit cycle oscillator models of the human circadian pacemaker. *Chronobiol Int* 22:613-629.
- Jenni OG, Achermann P, and Carskadon MA (2005) Homeostatic sleep regulation in adolescents. *Sleep* 28:1446-1454.
- Jewett ME, Forger DB, and Kronauer RE (1999). Revised limit cycle oscillator model of human circadian pacemaker. *J Biol Rhythms* 14:493-499.
- Kerkhof GA and Van Dongen HPA (1996) Morning-type and evening-type individuals differ in the phase position of their endogenous circadian oscillator. *Neurosci Lett* 218:153-156.
- Kerkhof GA (1985) Inter-individual differences in the human circadian system: A review. *Biol Psychol* 20:83-112.
- Klerman EB, Rimmer DW, Dijk DJ, Rizzo RE III, Kronauer JF, and Czeisler CA (1998) Nonphotic entrainment of the human circadian pacemaker. *Am J Physiol Regul Integr Comp Physiol* 274:R991-996.
- Klerman EB and Dijk DJ (2008). Age-related reduction in the maximal capacity for sleep—Implications for insomnia. *Curr Biol* 18:1118-1123.
- Kronauer RE, Czeisler CA, Pilato SF, Moore-Ede MC, and Weitzman ED (1982) Mathematical model of the human circadian system with two interacting oscillators. *Am J Physiol* 242:R3-R17.
- Kronauer RE, Forger DB, and Jewett ME (1999) Quantifying human circadian pacemaker response to brief, extended, and repeated light stimuli over the phototopic range. *J Biol Rhythms* 14:501-516.
- Leise T and Siegelmann H (2006) Dynamics of a multistage circadian system. *J Biol Rhythms* 21:314-323.
- Leloup JC and Goldbeter A (2003) Toward a detailed computational model for the mammalian circadian clock. *Proc Natl Acad Sci U S A* 100:7051-7056.
- Lu J, Shiromani P, and Saper CB (1999) Retinal input to the sleep-active ventrolateral preoptic nucleus in the rat. *Neuroscience* 93:209-214.
- Mallis MM, Mejdal S, Nguyen TT, and Dinges DF (2004) Summary of the key features of seven biomathematical models of human fatigue and performance. *Aviat Space Environ Med* 75:A4-A14.
- Mongrain V, Lavoie S, Selmaoui B, Paquet J, and Dumont M (2004) Phase relationships between sleep-wake cycle and underlying circadian rhythms in morningness-eveningness. *J Biol Rhythms* 19:248-257.
- Mongrain V, Paquet J, and Dumont M (2006) Contribution of the photoperiod at birth to the association between season of birth and diurnal preference. *Neurosci Lett* 406:113-116.

- Mongrain V and Dumont M (2007) Increased homeostatic response to behavioral sleep fragmentation in morning types compared to evening types. *Sleep* 30:773-780.
- Monk TH, Reynolds CF III, Buysse DJ, Hoch CC, Jarrett DB, Jennings JR, and Kupfer DJ (1991) Circadian characteristics of healthy 80-year-olds and their relationship to objectively recorded sleep. *J Gerontol* 46:M171-M175.
- Monk TH, Buysse DJ, Reynolds III CF, Kupfer DJ, and Houck PR (1995) Circadian temperature rhythms of older people. *Exp Gerontol* 30:455-474.
- Monk TH (2005) Aging human circadian rhythms: Conventional wisdom may not always be right. *J Biol Rhythms* 20:366-374.
- Münch M, Knoblauch V, Blatter K, Schröder C, Schnitzler C, Kräuchi K, Wirz-Justice A, and Cajochen C (2005) Age-related attenuation of the evening circadian arousal signal in humans. *Neurobiol Aging* 26:1307-1319.
- Nunez PL (1995) *Neocortical Dynamics and Human EEG Rhythms*. New York: Oxford University Press.
- Pace-Schott EF and Hobson JA (2002) The neurobiology of sleep: Genetics, cellular physiology and subcortical networks. *Nat Rev Neurosci* 3:591-605.
- Pandi-Perumal SR, Zisapel N, Srinivasan V, and Cardinali DP (2005) Melatonin and sleep in aging population. *Exp Gerontol* 40:911-925.
- Phillips AJK and Robinson PA (2007) A quantitative model of sleep-wake dynamics based on the physiology of the brainstem ascending arousal system. *J Biol Rhythms* 22:167-179.
- Phillips AJK and Robinson PA (2008) Sleep deprivation in a quantitative physiologically based model of the ascending arousal system. *J Theor Biol* 255:413-423.
- Porkka-Heiskanen T, Strecker RE, and McCarley RW (2000) Brain site-specificity of extracellular adenosine concentration changes during sleep deprivation and spontaneous sleep: An in vivo microdialysis study. *Neuroscience* 99:507-517.
- Reid KJ, Chang AM, Dubocovich ML, Turek FW, Takahashi JS, and Zee PC (2001) Familial advanced sleep phase syndrome. *Arch Neurol* 58:1089-1094.
- Rempe MJ, Best J, and Terman D (2010) A mathematical model of the sleep/wake cycle. *J Math Biol* 60:615-644.
- Robinson PA, Rennie CJ, and Wright JJ (1997) Propagation and stability of waves of electrical activity in the cerebral cortex. *Phys Rev E* 56:826-840.
- Roenneberg T, Kuehnle T, Pramstaller PP, Ricken J, Havel M, Guth A, and Mrosovsky M (2004) A marker for the end of adolescence. *Curr Biol* 14:R1038-R1039.
- Saper CB, Chou TC, and Scammell TE (2001) The sleep switch: Hypothalamic control of sleep and wakefulness. *Trends Neurosci* 24:726-731.
- Saper CB, Scammell TE, and Lu J (2005) Hypothalamic regulation of sleep and circadian rhythms. *Nature* 437:1257-1263.
- Scheer FA, Wright KP Jr, Kronauer RE, and Czeisler CA (2007) Plasticity of the intrinsic period of the human circadian timing system. *PLoS One* 2:e721.
- Schmidt C, Collette F, Leclercq Y, Sterpenich V, Vandewalle G, Berthomier P, Berthomier C, Phillips C, Tinguely G, Darsaud A, et al (2009) Homeostatic sleep pressure and responses to sustained attention in the suprachiasmatic area. *Science* 324:516-519.
- St. Hilaire MA, Klerman EB, Khalsa SBS, Wright KP Jr, Czeisler CA, and Kronauer RE (2007) Addition of a non-photic component to a light-based mathematical model of the human circadian pacemaker. *J Theor Biol* 247:583-599.
- Swaab DF, Chung WCJ, Kruijver FPM, Hofman MA, Hestiantoro A (2003) Sex differences in the hypothalamus in the different stages of human life. *Neurobiol Aging* 24:S1-S16.
- Swaab DF, Fliers E, and Partiman TS (1985) The suprachiasmatic nucleus of the human brain in relation to sex, age and senile dementia. *Brain Res* 342:37-44.
- Taillard J, Philip P, Coste O, Sagaspe P, and Bioulac B (2003) The circadian and homeostatic modulation of sleep pressure during wakefulness differs between morning and evening chronotypes. *J Sleep Res* 12:275-282.
- Tamakawa Y, Karashima A, Koyama Y, Katayama N, and Nakao M (2006) A quartet neural system model orchestrating sleep and wakefulness mechanisms. *J Neurophysiol* 95:2055-2069.
- Toh KL, Jones CR, He Y, Eide EJ, Hinz WA, Virshup DM, Ptáček LJ, and Fu YH (2001) An h Per2 phosphorylation site mutation in familial advanced sleep phase syndrome. *Science* 291:1040-1043.
- Tune GS (1968) Sleep and wakefulness in normal adult humans. *Br Med J* 2:269-271.
- Van Dongen HPA (2004) Comparison of mathematical model predictions to experimental data of fatigue and performance. *Aviat Space Environ Med* 75:A15-A36.
- Weitzman ED, Czeisler CA, Coleman RM, Spielman AJ, Zimmerman JC, Dement W, and Pollak CP (1981) Delayed sleep phase syndrome. A chronobiological disorder with sleep-onset insomnia. *Arch Gen Psychiatry* 38:737-746.
- Wittman M, Dinich J, Mrosovsky M, and Roenneberg T (2006) Social jetlag: Misalignment of biological and social time. *Chronobiol Int* 23:497-509.